

Global Facilities - Design & Construction Standard – Constructability Procedure

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[ECN History](#)[Submit a Change Request](#)

1.0 Introduction

- 1.1 Constructability is defined as “the optimum use of construction knowledge and experience in Design, planning, procurement, construction and operations to achieve overall project objectives.” The philosophies and principles of constructability are applicable to all Micron projects regardless of size and are based upon published recommendations of the Construction Industry Institute (CII) and established best practices of Micron, as well as other Owners and Contractor companies. The Micron Organization is committed to the effective implementation of a constructability process.
- 1.2 Constructability is a continuous improvement process. It is based on a forward-looking, integrated planning philosophy, in addition to backward-looking ‘review’ of completed designs, which will minimize design rework and result in higher quality projects. Activities, efforts, and results will be continuously evaluated, and associated processes and procedures will be updated, and improved.
- 1.3 The emphasis of this procedure is on the project level integration of construction expertise beginning very early in the front-end planning process (Appraise/Select), and continuing through field execution including commissioning, hook up, and start-up. The fundamental output of this ongoing effort is a “construction based” design and execution plan for the most efficient implementation of the work in the field.
- 1.4 Portions of this procedure have been italicized. This is to differentiate background or additional clarifying information provided to enhance the Project Management Team’s (PMT) understanding of certain elements from the specific procedural steps inherent to constructability implementation.

2.0 Scope

- 2.1 The scope of this procedure is to all Micron sites and targeted to all Micron team members working on construction projects within Micron sites.

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
CEP	Construction Execution Plans
CII	Construction Industry Institute
EHS	Environment, Health & Safety
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
PMT	Project Management Team
RFP	Request for Proposal

4.0 Roles and Responsibilities

4.1 Constructability Champion

- 4.1.1 Champion ongoing recognition/designation of a Micron constructability process
- 4.1.2 Garner management support for constructability implementation efforts/activities
- 4.1.3 Recognize barriers to constructability implementation and develop effective barrier breaking strategies
- 4.1.4 Manage the degree of constructability implementation documentation in place
- 4.1.5 Manage efforts to capture and communicate constructability lessons learned and success stories across all projects
- 4.1.6 Lead efforts to promote awareness and implementation of advanced construction technologies
- 4.1.7 Ensure that appropriate reference to constructability is contained in model contract documents
- 4.1.8 Provide general support to the Project Managers, and Project Constructability Coordinators
- 4.1.9 Lead efforts to track the value returned to Micron through the constructability process
- 4.1.10 Assist with ongoing implementation efforts and develop process updates
- 4.1.11 Recognize personnel and teams for their efforts in constructability implementation

4.2 Project Manager

- 4.2.1 Establish a project environment that is supportive of constructability implementation beginning very early in the project development process and continuing through field implementation
- 4.2.2 Emphasize the influence that project decisions have on the project's cost and insist that construction input be provided for the project's major decisions
- 4.2.3 Advocate the use of constructability to meet the traditional project objectives of EHS, quality, schedule, and cost
- 4.2.4 Provide an objective balance between both design and construction needs
- 4.2.5 Seek and gain funding and support for constructability implementation
- 4.2.6 Integrate key constructability activities and milestones into the project schedule
- 4.2.7 Integrate constructability into periodic project review agendas to heighten visibility
- 4.2.8 Provide timely recognition of constructability implementation efforts and results

4.3 PMO (with support from Project Manager)

- 4.3.1 Overall responsibility for the execution of this procedure over the entire life of the project
- 4.3.2 Designate the Project Constructability Coordinator (in consultation with the Project Manager and other key PMT members)
- 4.3.3 Include a Constructability Plan as part of the Construction Execution Plan
- 4.3.4 Establish project constructability implementation goals/objectives
- 4.3.5 Identify appropriate measures for constructability goals and objectives
- 4.3.6 Determine the level of formality for the constructability implementation
- 4.3.7 Consider constructability when selecting the contracting strategy
- 4.3.8 Select the organization members that will be responsible for constructability implementation
- 4.3.9 Consider constructability when developing the project contracting strategy
- 4.3.10 Guide the specific contract document language that will be employed to ensure adequate constructability implementation
- 4.3.11 Provide timely recognition of constructability implementation efforts and results

4.4 Project Constructability Coordinator (may also be the PMO)

4.4.1 Plan, implement, and monitor the project level Constructability program including:

- a) Organize the overall constructability plan and the timing
- b) Assemble, organize, and orient the team members
- c) Identify barriers and barrier breaking strategies that will be employed
- d) Coordinate review of lessons learned from past projects
- e) Plan and schedule Constructability workshop(s)
- f) Screen checklist ideas
- g) Ensure participation by appropriate ad hoc specialists
- h) Conduct the Constructability workshop(s)
- i) Identify constructability related concerns and opportunities, prioritize constructability concepts for implementation, and draft concept application plans (i.e. constructability studies) for needed deliverables that support critical decision
- j) Maintain the Constructability Suggestion Log for the duration of the project, tracking issues that have been considered, objective evaluation of benefits, and plans for/actual resolution including responsible parties
- k) Evaluate and report on constructability implementation progress and results, including effect on project budget and schedule
- l) Provide timely recognition of constructability implementation efforts and results
- m) Provide feedback to the Manager, Construction Services regarding potential upgrades to this Constructability Procedure

4.5 Project Constructability Team (including ad hoc Specialists)

4.5.1 This team is made up of representatives from all organizations supporting the project. This should include, but is not limited to, Construction, Engineering/Design, Functional Discipline Engineers, Procurement, Facilities, Commissioning, EHS, Quality Assurance/Control, Project Controls, Joint Venture Representatives, Contractors, and Sub-Contractors

4.5.2 Provide constructability input on a continuous basis

4.5.3 Develop benefits input and provide prioritization input

4.5.4 Develop constructability studies

4.5.5 Participate in constructability study implementation

5.0 Procedure

- 5.1 The constructability process at the project level (i.e. the program described in the following sections) consists of three critical activities:
 - 5.1.1 Define project constructability requirements, capabilities and framework.
 - 5.1.2 Plan project constructability implementation.
 - 5.1.3 Implement project constructability.
- 5.2 The constructability process should be implemented during initial phase (Appraise/Select) and continue through project planning, design, procurement, construction, commissioning, tool hook-up and start-up. The earlier the implementation in the project development process that the constructability program begins, the higher the potential savings. This will require the PMT assign Construction staff earlier in the project life-cycle than is traditional.
- 5.3 Constructability is not designed to be a burden to the project. It is intended to develop the smartest, most efficient work methods to meet the project's objectives. This requires combining the input of the Constructability Team using methodical communications that enable us to learn from each other and continuously improve our project performance.
- 5.4 The level of formality will differ depending on the project's size, cost, scope, contract strategy, and phase of the development. The constructability program will require leadership skills and initiative. The PMT Project Manager or one of the most experienced and knowledgeable direct reports must be chosen as the Constructability Coordinator. A Senior Engineer, PMO or CSA / MEP coordinator are prime candidates for the Constructability Coordinator role due to experience and because they typically supervise and coordinate the construction team and their work. The Project Management Team (PMT), associated support organizations, and the Contractors will be among the members of the Project Constructability Team.

6.0 Define Project Constructability Requirements, Capabilities and Framework

- 6.1 Define (Select) Constructability Team Members
 - 6.1.1 The Micron Project Manager and Constructability Team Members have a critical influence on the success of the project's constructability implementation efforts. The Project Manager must be committed to constructability, as this will have a direct effect on its effectiveness. The Project Manager must be able to establish a supportive project environment, commit to increased cost effectiveness, use constructability to meet other project objectives, and must involve construction resources early.
 - 6.1.2 Clear criteria should be established for the selection of Constructability Team Members. Selection criteria should include experience with other constructability implementation efforts, work experience, construction knowledge, communication skills,

teamwork skills, ability to objectively evaluate design and construction trade-off's, and receptiveness to new ideas and methods.

6.2 Define Constructability Objectives & Measures

6.2.1 Verify Project Objectives Considering Constructability

- a) Thoroughly educate all Constructability Team members of all of the project's objectives and resulting sub-objectives.
- b) The team should be operating with a very detailed list of project objectives and sub-objectives that are correctly prioritized. It is important to evaluate the design and construction implications when prioritizing the objectives. These objectives should be used later in the project to analyze trade-offs between alternatives influencing both design and construction.

6.2.2 Establish Constructability Objectives

- a) The Project Manager, PMO, and Constructability Coordinator should draft the constructability objectives within the context of the project objectives. These draft constructability objectives should be conveyed to and validated with the Constructability Team at the start of the Constructability Workshop. *In developing constructability objectives, consideration should be given to how objectives can be effectively measured. Constructability specific objectives may include items such as:*
 - EHS risk management
 - Development of an integrated engineering/construction plan
 - Inclusion of construction practice within the engineering/design phase
 - Inclusion of project specific construction methodology in the design (e.g. Space Management, cable routing, lateral designs, modular or packaging concepts, etc.)
 - Efficient procurement and contracting strategies
 - Standardized elements
 - Minimizing engineering rework
 - Use of modules/preassembly
 - Use of lifting/transportation equipment
 - Material lay-down areas
 - Ease of fabrication and construction
 - Use of physical models or CADD (BIM / 5D modelling)
 - Jobsite accessibility
 - Development of construction friendly specifications and drawings
 - Improved constructor/engineer communications
 - Minimizing construction rework
 - Minimizing jobsite congestion
 - Minimizing occurrence of labor disputes

6.2.3 Identify Appropriate Measures for Objectives

- a) Determine how each constructability objective will be measured. Cost and schedule performance are normally easily measured. Cost and savings estimates are commonly made for design alternatives and tradeoff analysis. Other possible performance measures include the following:
- Direct field labor hours
 - Labor productivity
 - Nonconforming items
 - Design rework hours
 - Change orders
 - Safety incidence probability
 - Cost of (transfer) work
 - Construction equipment cost

6.2.4 Determine Constructability Program Level of Formality

- a) Several methods exist to formalize a project constructability program. These can include preparing a budget, developing a Constructability Plan (could be part of the Construction Execution Plan), preparing procedures for major constructability tasks, issuing progress and status reports, tracking constructability suggestions, etc. *Each project must weigh the benefits of varying levels of program formality and tailor their program to meet the specific needs and characteristics of their project (e.g. construction type, contract type, project size, technical difficulty, Project geographical location, etc.).*
- b) All Construction Execution Plans (CEP) should include a Constructability Plan that is appropriate for the project scope. The Project Manager must communicate the fact that the cost influencing curve begins at project kickoff and is substantially complete at mobilization. With Constructability, early timing is of the essence.

6.2.5 Coordinate Constructability Framing with Selection of the Project Contracting Strategy

- a) The selection of the contract strategy has a profound impact on the timing and application of the constructability input. A contract strategy consists of as a minimum at least two parts:
- Contract arrangement
 - Contract type or commercial terms.
- b) In selecting a strategy Micron must perform the following:
- Assess the available in-house constructability expertise that can either lead or enhance the project constructability effort.
 - Identify available Micron constructability resources. The PMT should identify and assess existing in-house *resources available to lead, participate, or consult on constructability.*

- *If prior formalized constructability implementations have occurred, the PMT should converse with those PMT's to understand the prior program details, who participated, get recommended participants for their implementation, and gather any pertinent lessons learned.*
- *Seeking out experienced Micron resources first will allow for continued consistent growth of a culture of constructability as a key work process.*
- Recognize how the selection of a contracting strategy impacts project constructability.
 - *Contract strategy can have an impact to constructability. Regardless of which contracting strategy is selected, the PMT should keep the ability to effectively accomplish constructability as a very high priority project objective.*
 - *Strategies such as combined EPC can make competent contractor personnel easily available to participate in constructability implementation.*
 - *Other strategies resulting in the retention of construction contractors or construction management firms later in the project development process will require the employment of alternative means to ensure that competent constructability expertise is available to the team.*
- Select the best contractor who is also suited to be responsible for constructability implementation during the initial stages of project definition.
 - *Once a contracting strategy is selected, Micron should formally decide which organization (Micron, constructor, design / PMO, etc.) is responsible for implementing and monitoring the constructability program.*
 - *Micron implemented programs are generally more beneficial for transferring lessons learned across project boundaries.*
 - *Other responsibility scenarios must be carefully considered in terms of contract language, program specific requirements, expertise to be employed, deliverables required, etc.*

6.2.6 Secure Contractors, Vendors & Consultants

- a) Once the contracting strategy has been selected, the next step is to determine how to facilitate early implementation of the constructability process. When the construction contractor or Project Manager has not been selected early enough to contribute to the high cost impact decisions made during the conceptual planning phase of a project, a surrogate may be required to provide the necessary construction knowledge and experience input. The surrogate may come from Micron, a

design organization, a constructor, a PMO, or a consultant. The surrogate's input shall be reviewed by the Micron Construction Services Department.

- Require a constructability process as part of tender documents, contractor prequalification, and ensure clarity of expectations for the effective, project specific implementation of a constructability program.
- When the contracting strategy involves early selection of the constructor or requires the utilization of other resources, Micron should include in their contractor prequalification process criteria that includes utilization and effectiveness of the contractor's constructability process (i.e. bridging from this Procedure to the contractor's procedures).
- Factors to consider include the following:
 - *Assessment of the contractor's corporate philosophy and recognition of the value of constructability implementation and its integration into their work processes*
 - *Evaluation of the contractor's documentation including existing manuals, methods used to track progress, and incorporation of lessons learned*
 - *The contractor's breadth of experience in implementing their constructability process and the availability of their personnel experienced in their process*
 - *Evaluation of the contractor's tools used to implement constructability, including how they evaluate and implement new technology*
 - *Use of Micron Request for Proposal (RFP) or project specifications to obtain constructability*
 - *When Micron specifies constructability requirements in their bid documents, a wide variation in the level of detail can and does occur. Several tools are available in the **Construction Industry Institute, Constructability Implementation Guide, Second Edition** that can assist with a thorough definition of constructability process implementation requirements.*

6.2.7 Consider incentive clauses

- a) Well-conceived incentive clauses may effectively enhance a project's constructability program.
- b) In many cases, other companies have used complementary incentive schemes shared between, or influenced by, both the designer and the constructor. Establishing common goals can ensure that shared objectives can be effectively met.

7.0 Plan Project Constructability Implementation

7.1 The following steps have been identified to lead to effective planning for constructability implementation

7.1.1 Develop the constructability team and identify/address barriers

7.1.2 Screen past lessons learned

7.1.3 Conduct the constructability planning workshop

7.1.4 Draft constructability studies.

- a) Timely and thoughtful planning is critical to effective constructability implementation.
- b) Planning should begin immediately upon formation of the PMT.
- c) Constructability planning efforts should also involve all major project stakeholders and team participants in a timely manner, exploiting each beneficial window of opportunity that can positively affect supportive decision making.
- d) CII defines 16 constructability concepts (premier guiding principles, see **Appendix A**) designed to be implemented during various phases of project development. These can be considered opportunities for implementation.
- e) Micron has developed a collection of discipline specific checklists (see **Appendix J**) containing specific constructability tactics.
- f) These checklists are Microns current chosen method of capturing lessons learned from implementations on an ongoing basis (i.e. periodic checklist revisions).

7.2 Team Development & Barrier Removal

7.2.1 To ensure mutual interests and to minimize any adversarial relationships, the following four tasks should be conducted regardless of project size. These tasks can be the focus of a Constructability Kick-Off Workshop (*i.e. an alignment workshop*) prior to implementing the first Constructability Planning Workshop.

- a) Assemble and organize the constructability team
 - The organizational structure of the constructability team can vary from project to project. A simple chart or table should be employed to delineate their roles.
 - The development of the team(s) does not need to be a lengthy process that is accompanied by complex flow charts and very detailed organizational charts demonstrating the interaction.
 - A methodical approach is required to divide the job into sections for improvement opportunities. An example chart and a table format are included as **Appendix B** and **Appendix C** respectively.
- b) Orient the constructability team
 - The Project Manager should convey to the constructability team four very important tenets that are key to the success of

the constructability implementation:

- Definition of constructability
 - That constructability will be used as a tool to enhance achievement of the project objectives. The team should also refer to previously developed constructability objectives and measures as this will help facilitate team commitment
 - A list of the project objectives that will be enhanced through constructability implementation
 - An emphasis on the need for teamwork
- c) The members of the constructability team may require some degree of training to be effective. Training topics could include the following:
- Definition of constructability
 - Constructability plan for the project
 - Importance of teamwork and effective communication
 - Project objectives and status
 - Project critical success factors
 - Review of constructability concepts and/or tactics checklists
 - Roles and responsibilities of constructability team members
 - Understanding of how constructability affects each team member (i.e. "how does this affect my job")
- d) Perform team building exercises
- A constructability program is more successful if the constructability team has unified project objectives and its members are communicating openly.
 - Specific attention must be given to developing a cohesive team.
 - Team building exercises can enhance and hasten the team building process.
 - Once the team is formed, the Constructability Coordinator should periodically evaluate its effectiveness.
 - Some large projects have prepared teamworking guides that address topics such as the following:
 - The project's concept of teamwork
 - Definition of team
 - Characteristics of high performing teams
 - Four natural stages of team development (i.e. orientation, dissatisfaction, resolution, and production)
 - Skills necessary to help teams develop
 - Methods/tools to facilitate teamwork
 - Implementation plan for building teamwork into constructability
 - Additional team building resources.

- e) Identify barriers present and define barrier breaking strategies
 - A wide variety of barriers makes constructability implementation more difficult. Some barriers relate to Micron commitment to undertake a constructability process: lack of constructability awareness, reluctance to provide front-end funding, and the perception that constructability is already being done effectively. Other barriers are related to team members' unfamiliarity with the philosophy and workings of the constructability process. **Appendix D** defines a number of potential barriers and strategies for addressing them.
 - One tactic that can be employed to reveal barriers is to have team members participate in a barrier identification exercise. Members of the constructability team should then participate in an exercise to identify the means by which they will overcome the barriers. **Appendix D** also contains a sample worksheet and chart that can be utilized to graphically display the results of barrier identification exercise(s) for communication purposes.

7.3 Screen Past Lessons Learned

- 7.3.1 Constructability teams are most effective when they can readily exploit the lessons and innovations experienced on past projects.
- 7.3.2 Effort should be made to gather as much recent past project information as possible for the team to review.
- 7.3.3 Routine update of portions of this Procedure will also serve to incorporate constructability lessons learned and make them routinely available to all PMTs.

7.4 Conduct the Constructability Planning Workshop

- 7.4.1 Prepare for and conduct the planning workshop
 - a) The workshop should be conducted with all key stakeholders and constructability team members in attendance and should be scheduled and implemented no later than DR 1.
 - b) The chief purpose of the constructability planning workshop is to identify constructability related concerns and opportunities, to prioritize those items, and to develop Constructability Studies for their execution within the context of the planned project progress.
 - c) The expected outcomes of the workshop are ready-to-implement studies that will facilitate effective and timely decision making supportive of project and constructability objectives. **Appendix E** is a sample agenda noting the activities that should be included in the workshop.
 - d) In planning the workshop, the "CII Constructability Application Matrix for Industrial Projects", included as **Appendix A**, may prove useful.
 - e) This matrix defines the notional timing relationships between

the 16 constructability concepts mentioned in section 6.2 and the project phase.

- f) Additionally, **Appendix J** is Discipline Specific Tactical Constructability Checklists that are compilations of issues from a variety of sources (e.g. Micron lessons learned and experiences, design and construction companies, etc.).
- g) These checklists are good for stimulating conversation and causing the team to think about details learned over many years of execution. These checklists are also the key part of this Procedure that will be periodically updated as new lessons are learned so that they are readily available to all PMT's engaged in constructability.
- h) As constructability ideas are generated, they shall be tracked on a Constructability Suggestion Tracking Log similar to that shown in **Appendix F**.
- i) This log will be a key document to track, at a high level, all constructability suggestions, Cost and schedule impacts (or other project objective impacts), decisions around implementation, and status.
- j) This log should be used as a source of information for generating constructability implementation reports showing items such as:
 - Total number of suggestions this period (PMT should determine the reporting period)
 - Total number of suggestions for the project
 - Total value of all suggestions
 - Total value of all suggestions approved
 - Total value of suggestions completed
- k) The Constructability Suggestion Tracking Log will be a key mechanism by which the Micron Organization will monitor construction preparedness in terms of creating an optimized construction-based approach to execution.
- l) Effective constructability implementation, as evidenced by a well-kept log with effective resolutions to all items, will develop the belief that the project's construction plan is well thought out and can be implemented with reduced risk.
- m) There are many ways the Constructability Planning Workshop can be formatted. Another approach is to focus first on the major construction methods linked to a complementary contracting strategy such as:
 - Modularizing process / mechanical units and pipe racks.
 - Preassembling steel, pipe spooling mechanical packaging vs. stick-building
 - Pre-casting select concrete structures vs. cast-in-place

- Fireproofing the steel at the shop vs. field
 - Mass excavation, bath tub excavation, or individual excavation
 - Activities affected by weather conditions
 - This could be followed up with a use of the Discipline Specific Tactical Constructability Checklists to develop more detail, priorities and finish with the development of Constructability Studies.
- n) While conducting the Constructability Planning Workshop, the Constructability Coordinator and PMT may want to consider combining the following activities which have some common purposes with constructability implementation, and can generate valuable ideas at a minimal cost:
- Contractor and Supplier lunch and learn sessions
 - Value Improving Practice Workshops
 - Reviewing CII research briefings and publications
 - Reviewing specifications and standards for improvements
 - Engineering and model reviews
 - Town Hall Meetings for reward and recognition
 - Knowledge Sharing Networks of Excellence (NoE) for locating expertise and reference materials

7.5 Draft Constructability Studies

- 7.5.1 The primary output of the Constructability Planning Workshop is to identify the required Constructability Studies that will elaborate on needed implementation deliverables. **Appendix G** is an example Constructability Study format that can be used to document plans for individual or groups of identified issues and opportunities to be completed post-workshop.
- 7.5.2 The intent is to positively impact decision making that supports effective project execution.
- 7.5.3 Some examples of decision-oriented studies that might be identified through this process include:
- a) Listing of modularization opportunities, including the scoping of the packaged units
 - b) Decision analysis for choice of structural system material/configuration
 - c) Preparation and analysis of a start-up and construction sensitive milestone schedule
 - d) Analysis of heavy lifts, including (but not limited to) crane selection, lift sequencing, and crane siting and accessibility
 - e) Logistics routing to ensure optimal delivery of equipment and materials
 - f) Preparation of a plan for temporary construction site facilities including warehousing and preservation of equipment and

materials

- g) Identification of technical specifications that require creation/update
- h) Analysis of site accessibility challenges and identification of related optional solutions
- i) Identification of options for maximizing field work progress

7.5.4 *Each decision or grouping of related decisions should be supported with a Constructability Study.* For smaller projects, or for those situations when the workshop duration can be extended over several days, workshop breakout groups can initiate the drafting of actual deliverables described in each Constructability Study. In most cases, such drafts will continue to be refined and completed over the appropriate "opportunistic schedule windows."

7.6 Finalize Constructability Studies

7.6.1 Constructability ideas need to be planned for application during the conceptual planning, design, and construction phases of projects. The Constructability Coordinator needs to coordinate constructability studies and integrate constructability activities into the project schedule so that adequate time is allowed for all constructability activities and so that design rework is not necessary.

8.0 Implement Project Constructability

8.1 The following steps have been identified for implementing constructability

- 8.1.1 Perform constructability studies
- 8.1.2 Monitor/evaluate constructability effectiveness
- 8.1.3 Document lessons learned.

8.2 Perform Constructability Studies

- 8.2.1 The constructability activities detailed in the Constructability Studies are a series of detailed actions that should be performed in a timely manner.
- 8.2.2 These actions should be implemented appropriately through all phases of project development. Ongoing constructability feedback should continue through project closeout, though the degree of formality can change as project execution progresses.
- 8.2.3 The Constructability Coordinator should monitor the implementation of constructability activities and take corrective action as appropriate.
- 8.2.4 Some additional considerations for constructability implementation are as follows:
 - a) The constructability team should convene on a regular basis to discuss constructability concepts, update status of constructability studies, track implementation of activities defined as part of the studies, document lessons learned and

provide constructability input as designs are produced.

- b) Particular emphasis will be placed on certain team members to provide constructability expertise when their specific area is being considered.
- c) Most trade-off analyses and comparisons will be based on rough order of magnitude, cost and schedule knowledge of the participants involved. This saves time and accelerates decisions. Professional estimating and scheduling support may be required for complex analyses.
- d) The Constructability Coordinator should monitor and periodically report on progress of the constructability implementation.
- e) At the appropriate time prior to release of a design package, the constructability team may perform a final constructability review to check for completeness and accuracy of design details. This review is not intended as an opportunity to modify the design, rather it will be a final check that approved concepts have been incorporated.
- f) During the construction phase, an active dialogue should be maintained with the contractor(s) to evaluate constructability items implemented and account for areas of possible improvement on future projects. These lessons learned may then be incorporated into this Procedure.
- g) As project development progresses, new constructability inputs can be received from within the Constructability Team as well as from personnel and organizations outside the team.

Appendix H is a Constructability Suggestion Form to provide constructability input for consideration. The suggestion should also be entered on the Constructability Log for tracking purposes. If the item is significant enough to gain priority status, it should be developed through a Constructability Study and implemented alongside all others.

8.3 Monitor/Evaluate Constructability Effectiveness

- 8.3.1 Tangible benefits be they financial, schedule, quality, safety, operational, or other, should be made available through periodic reports outlining the progress of the project's constructability implementation and how it is impacting the project's objectives. Reports may also address implementation progress.
- 8.3.2 At the conclusion of the project, the Constructability Team should jointly assess the project constructability program performance and focus on areas for improvement.
- 8.3.3 Periodically the Micron Constructability process will be assessed by the Constructability Champion for effectiveness based on the following criteria:
 - a) Are overall process goals and objectives being met?

- b) Is there a need to establish new process goals?
- c) Do project level constructability processes receive all the necessary support to be successful?
- d) Are barriers to constructability implementation being effectively removed?
- e) Are training programs including high value awareness materials?
- f) Are lessons learned being effectively cross-communicated (i.e. procedure and work process revision)?
- g) Are there lessons learned in the methodologies by which trade-off evaluations are being made?
- h) Is the process receiving adequate visibility?
- i) Are contributors being appropriately recognized for their contributions?

8.3.4 **Appendix I** is a Constructability Program Evaluation Matrix that will be used to periodically assist with the above assessment. Periodically, PMT's that have recently completed constructability implementation will be asked to assist with the process evaluation.

8.4 Document Lessons Learned

- 8.4.1 Feedback on project constructability implementation shall be reported periodically to the Project Manager, to allow adjustments to this Procedure.
- 8.4.2 It will be important to request a specific assessment of the design documents from the constructor(s).
- 8.4.3 Lessons learned should be documented throughout project execution rather than just at project closeout.

9.0 Document Control

9.1 Approval: FCG_ENG_CONSTRUCTION_APPR

9.2 Notification:

- GLOBAL_EHS
- GLOBAL_EHS_SAFETYHEALTH
- GLOBAL_FAC_COMMUNICATIONS
- GLOBAL_FAC_CONST_SOP
- GLOBAL_FAC_MANAGERS
- GLOBAL_FAC_NOTIFY
- GLOBAL_SECURITY_TEAM
- GP_INDIRECT_FACILITIES
- FCG_F4_FAC_NOTIFY
- FCG_F6_FAC_NOTIFY
- FCG_F10_FAC_NOTIFY
- FCG_F11_FAC_NOTIFY
- FCG_F15_FAC_NOTIFY
- FCG_F16_FAC_NOTIFY
- FCG_MMP_FAC_NOTIFY
- FCG_MMY_FAC_NOTIFY
- FCG_MSB_FAC_NOTIFY
- FCG_MTB_FAC_NOTIFY
- FCG_MXA_FAC_NOTIFY
- WW_FACILITIES_MANAGERS_APPR

9.3 Final Viewer Access:

All Micron sites Facilities Team Members

9.4 Retention: Adherence to the requirements specified in the [Global Facilities - Document Control Procedures](#)

9.5 Review: Biennial

10.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	N/A	New Document	11/26/2019
1	Title Section 9	Changed document title from Global Facilities - Constructability Procedure -XXX - SITE to Global Facilities - Design & Construction Standard - Constructability Procedure Updated Approval Group to FCG_ENG_CONSTRUCTION_APPR Updated the Notification Groups to GLOBAL_EHS, GLOBAL_EHS_SAFETYHEALTH, GLOBAL_FAC_COMMUNICATIONS, GLOBAL_FAC_CONST_SOP, GLOBAL_FAC_MANAGERS, GLOBAL_FAC_NOTIFY, GLOBAL_SECURITY_TEAM, GP_INDIRECT_FACILITIES, FCG_F4_FAC_NOTIFY, FCG_F6_FAC_NOTIFY, FCG_F10_FAC_NOTIFY, FCG_F11_FAC_NOTIFY, FCG_F15_FAC_NOTIFY, FCG_F16_FAC_NOTIFY, FCG_MMP_FAC_NOTIFY, FCG_MMY_FAC_NOTIFY, FCG_MSB_FAC_NOTIFY, FCG_MTB_FAC_NOTIFY, FCG_MXA_FAC_NOTIFY, WW_FACILITIES_MANAGERS_APPR	11/24/2021

11.0 Appendices

S/N	Appendix	Description
1	Appendix A	Concepts Application Matrix for Industrial Projects
2	Appendix B	Constructability Team Sample Organization Chart
3	Appendix C	Constructability Team Sample Organization Table
4	Appendix D(a)	Constructability Barriers Checklist (Explanation)
5	Appendix D(b)	Constructability Barriers Checklist
6	Appendix D(c)	Constructability Barrier Scoring Worksheet & Assessment Results
7	Appendix D(d)	Constructability Barrier Breakers (Explanation)
8	Appendix D(e)	Constructability Barrier Breakers Checklist
9	Appendix E	Constructability Planning Workshop Agenda
10	Appendix F	Constructability Suggestion Tracking Log
11	Appendix G	Constructability Study
12	Appendix H	Constructability Suggestion Form
13	Appendix I	Constructability Process Evaluation Matrix
14	Appendix J	Tactical Constructability Checklists (Tabbed Excel Workbook)
15	Appendix K	Flowchart of Activities in Constructability Procedure

APPENDIX A: CONSTRUCTABILITY CONCEPTS APPLICATION MATRIX FOR INDUSTRIAL PROJECTS

No.	Concept Description
I-1	Constructability implementation plans are an integral part of the <i>Project Execution Plan</i> .
I-2	Early project feasibility planning takes advantage of construction knowledge and experience.
I-3	Development of the project contracting strategy involves construction knowledge and experience.
I-4	Project schedules are construction- and startup-sensitive.
I-5	Important, early design decisions consider modularization / preassembly, construction automation, and other major construction method options.
I-6	Permanent and temporary site layouts promote efficient construction.
I-7	Advanced information technologies are applied to facilitate efficient construction.
II-1	Design and procurement schedules are construction-sensitive.
II-2	Designs are configured to enable efficient construction and use of efficient technologies.
II-3	Design elements are standardized.
II-4	Procurement, construction and startup efficiency are considered in the development of contract documents.
II-5	Module / preassembly designs facilitate fabrication, transport, and field installation.
II-6	Designs promote construction accessibility of personnel, material, and equipment.
II-7	Designs facilitate construction and field productivity under adverse weather conditions.
II-8	Project plans enhance security during construction.
III-1	Innovative construction management and field methods are applied to increase construction efficiency.

Conceptual Planning Phase

I-1 – Constructability implementation plans are an integral part of the Project Execution Plan

If constructability is to be effectively achieved on a project, the plan for achieving it must be written as part of the project execution plan. The constructability program should become an integral part of the planning process for the project. The constructability program can contribute to effective project execution in many ways, including: (1) helping to establish project goals and objectives, (2) providing a logical and systematic manner for integrating design and construction, (3) providing a mechanism for obtaining field construction experience as needed, and (4) improving the understanding of the design intent by construction personnel.

I-2 –Early project feasibility planning takes advantage of construction knowledge and experience

The inclusion of construction personnel in early planning teams can be critical to achieving cost and schedule benefits. For example, construction knowledge and experience on early project planning teams can be actively involved in a variety of critical decisions or assessments: (1) establishing project objectives, (2) selecting project site, (3) analyzing schedule feasibility, (4) establishing productivity assumptions, (5) preparing estimates and budgets, and (6) identifying sources of long-lead and / or scarce materials and equipment

I-3 – Development of the project contracting strategy involves construction knowledge and experience

The contracting strategy will have a major influence on the availability of qualified construction personnel to serve on constructability teams. Micron must be particularly cognizant of strategies that limit the role of the constructor during early phases of the project. For example, if the project delivery system employs the traditional strategy of separate contracts for the designer and general contractor, Micron may have primary responsibility for coordinating early construction input. Special arrangements, such as use of a constructability consultant, may be required to obtain the necessary constructability input during the conceptual and early design phases. Regardless of which approach is selected, the appropriate construction personnel must be involved, and their expected qualifications must be clearly specified in the contract documents.

I-4 –project schedules are construction and startup sensitive

This concept establishes the principle that the project completion date and the requirements of the construction and startup phases should be considered in optimizing the overall project schedules. The overall project schedule, in allocating durations of time among the major project activities should achieve an optimal economic balance. To reap the full benefits of constructability, a “backward pass,” “feed-forward,” or construction-driven approach should be applied in developing the master schedule. Of course, while using this approach, a proper balance must be sought among the times allocated for planning, design, procurement, construction, and startup.

I-5 –Important, early design decisions consider modularization / preassembly, construction automation, and other major construction method options

Major construction methods can be defined as the use of construction equipment, labor, and work sequencing in such a way that the methods become a major design “driver.” Thus, a design-driver is a construction method, condition, or technique that project designers must address or accommodate, and which if substituted later in project, could significantly impact cost, time, or performance results. Major construction methods must be considered during conceptual planning. All members of the project team should interact and contribute to the decision for selecting such major methods. For many projects, perhaps there is no method selection option more critical in this respect than that of scoping modularization and major pre-assembly efforts. In addition, as construction automation services begin to play a larger role in project execution, associated design requirements must be considered in a timely manner for the technologies to deliver their full benefit.

I-6 – Permanent and temporary site layouts promote efficient construction

Construction efficiency is an important criterion in the layout of both permanent and temporary facilities. Permanent facility layout should be reflective of construction concerns and well-coordinated with temporary facility plans. Construction considerations that should be provided for include:

- (1) adequate space for lay-down and fabrication areas
- (2) access for construction equipment, materials, and personnel
- (3) avoidance of costly and problematic types of construction such as underground and elevated work when alternatives exist
- (4) temporary construction uses of existing or permanent facilities
- (5) planning for adequate drainage during all phases of construction.

I-7 – Advanced information technologies are applied to facilitate efficient construction

Constructability can be enhanced by exploiting the capabilities and benefits of advanced information technologies, which are ever-emerging. The use of advanced technologies has the potential to revolutionize the methods and processes used by the construction industry. Some of the information technologies that are benefiting the constructability of projects include the use of “5D” computer modeling for avoidance of physical interferences and “walk-thru” or construction simulation; relational database systems for lessons-learned systems; work process simulation for optimizing resource utilization; web-based team collaboration systems for enhancing communications and information access; bar coding of materials, equipment and workers; field notebook computers to facilitate progress tracking and inspections; and remote sensing technologies for verifying field dimensions; among others. Beyond enabling beneficial data manipulation, such technologies provide opportunities to more effectively apply construction knowledge and experience through enhanced interfaces between project engineering, construction, and maintenance personnel.

Design and Procurement Phase**II-1 – Design and procurement schedules are construction-sensitive**

Construction normally is a large cost segment of a project and thus can exert great cost leverage on a “backward pass” schedule through both procurement and detailed design. Procurement processes, other than for lead times, may have only a moderate ability to conform to the needs of construction processes. While design is a complex and demanding process, it frequently is a fruitful area for optimizing the entire project schedule. Of course, Micron should be sensitive to the cost and schedule trade-offs that exist over the entire period of project activity.

II-2 – Designs are configured, and permanent equipment is selected to enable efficient construction and use of efficient technologies

In order to achieve optimal design configurations, ideas need to be exchanged between construction and design professionals before “pencil on paper” design activities occur. Designers should be mindful of such factors as configuration simplicity, minimal number of parts, field flexibility, intuitive methods for alignment and positioning, tool accessibility, reduction of skill-intensive or resource-limited operations, etc. Many of these same principles apply in the selection of engineered equipment. Noteworthy, as with several other concepts, in order to achieve these goals, organizations need to make special efforts to remain up-to-date on current technologies.

II-3 – Design elements are standardized

This concept addresses the achievement of cost and schedule benefits through the use of Standardization, which, broadly applied, is a process by which designers seek project elements that are both regularly and widely used, are available, or are readily supplied. Usually the predominant trade-off is a cost reduction that results from time savings in construction and volume purchase discounts versus an increase in materials due to a more conservative design. Additional advantages of standardization include simplified material procurement and materials management, reduced design time, and greater interchangeability of spare parts during maintenance operations.

II-4 – Procurement, construction and startup efficiencies are considered in the development of contract documents

Construction knowledge and experience can contribute significantly to the generation of specifications and drawings that promote efficiency in field construction operations. Constructability can be enhanced when underlying corporate guide specifications offer clear-cut options, when specifications development occurs with full and early involvement of personnel with appropriate construction knowledge and experience, and when there is sufficient time to develop complete, consistent, and error-free specifications. Additional effective practices include maintaining specifications so that the most current cost-effective state-of-the-practice techniques and materials are specified and avoiding those approaches where a single item is specified multiple times and “order of precedent” must be routinely examined in order to determine applicability. Also, the cost saving potential of “or equal” specifications should be balanced against the management burden and risks involved. Likewise, project drawings can be prepared in ways that greatly enhance field productivity. As an example, some have found that separating concrete dimensions for formwork from those for reinforcing steel on two different drawings can result in significant field productivity gains. As for Startup, planning and field efforts are greatly facilitated by adding startup system numbers within the title block of all design drawings.

II-5 – Module / preassembly designs facilitate fabrication, transport, and field installation

Once decisions have been made to modularize or preassemble major project components, special factors must be addressed during design and procurement to ensure successful implementation. Designers should first consider where fabrication will occur and when the vendor’s shop is under controlled conditions, tolerances can be made more stringent. Also, module / preassembly designs add requirements for transporting and handling large assemblies and raise questions about equipment availability, sequence, and access. Finally, module site installation methods must be considered early because they affect module design, overall plot layout, and design of services, foundation supports, and inter-connecting systems.

II-6 – Designs promote construction accessibility of personnel, material, and equipment

Difficult access for personnel, material, and equipment on a project can negatively impact the achievement of project objectives. Difficult access for personnel can severely impact productivity and difficult access routes frequently present unsafe working conditions. Similarly, difficult access routes for high volume commodity materials can adversely impact cost and schedule. Accessibility studies for major pieces of equipment and temporary erection access openings are as important as those for personnel and commodities. For example, specific issues that should be considered include:

- (1) sequencing of work
- (2) delivery schedules for major pieces of equipment
- (3) lay-down areas on congested sites
- (4) delivery routes
- (5) use of permanent elevators for personnel lifts
- (6) installation and location of underground work to be traversed later by heavy equipment
- (7) type, location, and required opening size for equipment.

II-7 – Designs facilitate construction and field productivity under adverse weather conditions

Adverse weather conditions can have significant impacts on construction cost and schedule. In many cases, however, the effects of adverse weather can be alleviated with proper considerations during design, such as the following:

- (1) Planning an accessible site layout
- (2) Providing protection to personnel
- (3) Judicious scheduling of design
- (6) Planning construction lighting
- (7) Planning for adequate site drainage
- (8) Scheduling and controlling delivery of equipment and materials to avoid unnecessary protection requirements.
- (9) Providing adequate temporary protected storage areas.

II-8 – Project plans enhance security during construction

In this era, it is particularly important that capital projects be secure to enable effective and safe project progress during construction and safe and dependable facility operations after project completion. In order to accomplish this, project planners should examine opportunities during planning, design, and procurement that can lead to a more secure site and that can ensure secure field progress and related work processes. Planning must include site access control, personnel surety investigations and control, and attention to document and information technology security. Appropriate security measures will be dependent on the results of a vulnerability assessment of specific project circumstances. Examples of related tactics include careful site selection, “hardening” the design, provisions for temporary protection of vulnerable features or project components, tamper-proof details where critical, perimeter security during construction and specifying security checks during off-site fabrication and transport.

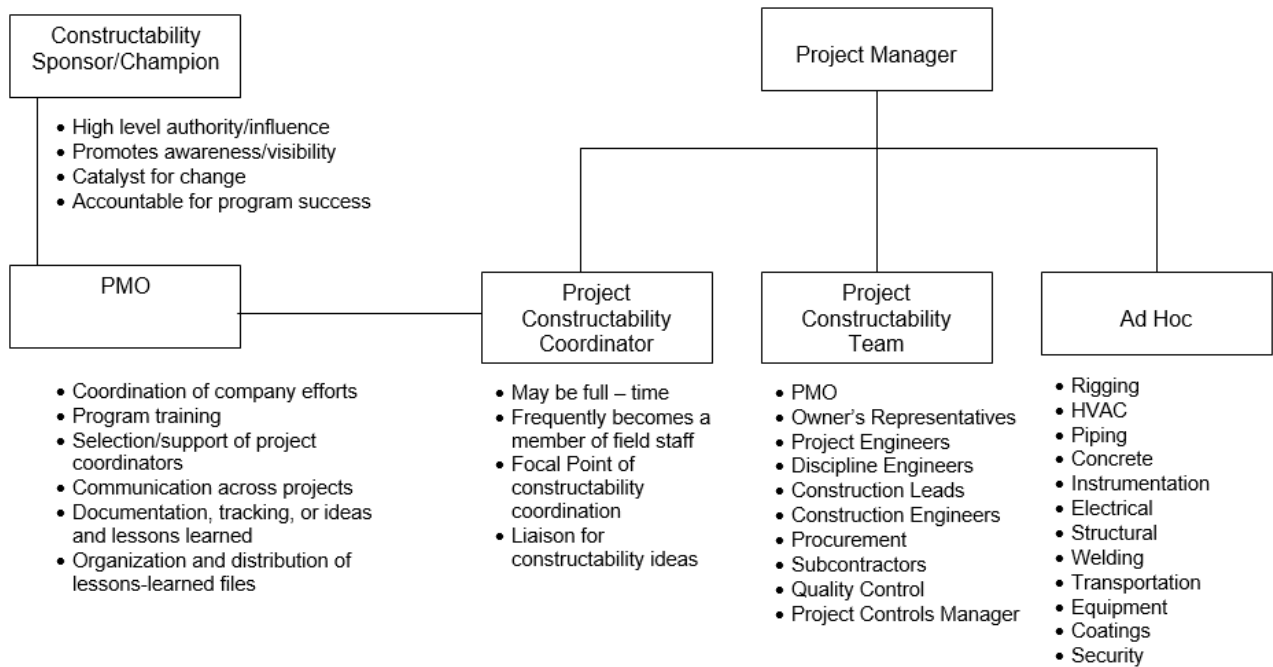
Field Operations

III-1 –Innovative construction management and field methods are applied to increase construction efficiency.

There is also a need to apply construction knowledge and experience to improve the effectiveness of site operations. Construction method innovations are typically “small” advances yet can be numerous. Thus, they should not be overlooked. Collectively, the potential benefit can be substantial. Innovative construction methods may “come in many forms”:

- (1) The sequencing of specific tasks
- (2) Use of temporary construction materials / systems
- (3) Use of specialized tools / equipment
- (4) Use of specialized construction equipment
- (5) Constructor-optional preassembly
- (6) Post-bid constructor preferences relevant to the layout, design, and selection of permanent materials. To be most successful in applying this concept, contractors must become “learning organizations” that promote “bottom-up” innovations and go a step further in communicating such innovations.

APPENDIX B:
CONSTRUCTABILITY TEAM SAMPLE ORGANIZATION CHART



APPENDIX C: CONSTRUCTABILITY PROCEDURE TEAM – SAMPLE MEMBERSHIPS

TEAM	TEAM	TEAM	TEAM	TEAM	TEAM	TEAM	TEAM
CSA	Mechanical	E & I & C	Construction	Gas	Chemicals	H/U	BIM
Scope:	Scope:	Scope:	Scope:	Scope:	Scope:	Scope:	Scope:
Demolition	Mechanical Eq	Electrical	Temporary Facilities	Base Build	Base Build	Temporary Facilities	Space Management
Site Work	Piping	Instrumentation	Temporary Services	Temporary Services	Temporary Services	Temporary Services	Model
Concrete	Process	Control Systems	Equipment & Tools	Equipment & Tools	Equipment & Tools	Equipment & Tools	
Steel	Fireproof		Staffing				
Buildings	Insulation						
Members:	Members:	Members:	Members:	Members:	Members:	Members:	Members:
Engineering	Engineering	Engineering	Engineering	Engineering	Engineering	Engineering	Engineering
Procurement	Procurement	Procurement	Procurement	Procurement	Procurement	Procurement	Procurement
Construction	Construction	Construction	Construction	Construction	Construction	Construction	Construction
EHS	EHS	EHS	EHS	EHS	EHS	EHS	EHS
Contracts	Contracts	Contracts	Contracts	Contracts	Contracts	Contracts	Contracts
Controls	Controls	Controls	Controls	Controls	Controls	Controls	Controls
Quality	Quality	Quality	Quality	Quality	Quality	Quality	Quality
Comms	Comms	Comms	Comms	Comms	Comms	Comms	Comms
Facilities	Facilities	Facilities	Facilities	Facilities	Facilities	Facilities	Facilities

APPENDIX D (a): CONSTRUCTABILITY BARRIERS CHECKLIST

For a constructability process to be effective, barriers must be recognized and overcome. This Tool is a checklist for determining the prevalent barriers identified that affect a corporate constructability process.

For the initial constructability kick-off workshop responses should be based on experiences on prior projects. Re-evaluation at subsequent workshops can then focus on recent experiences on the current project. The request for response should be framed as, "could the barrier be significant on the current project if it is not addressed."

To perform the self-assessment, distribute the Barriers Checklist to a variety of personnel; multiple assessments from varied levels within the entire participant group must be solicited for the evaluation to be effective. Each participant should determine whether each particular barrier is insignificant, significant, or very significant and needs to be addressed and overcome.

This checklist is not a comprehensive listing of all potential barriers; other barriers may exist within a particular organization or a specific project. Respondents should not hesitate to identify additional barriers they feel warrant attention.

Results from these assessments should be captured and can be assigned scores via a simple weighting system such as one (1) point for insignificant, three (3) points for significant, and five (5) points for very significant responses. An Example worksheet and chart is included as **Appendix D** to allow capture and depiction of the results of the assessment.

The Barriers Checklists should be periodically re-evaluated to determine if those barriers that were identified have been mitigated, or if new barriers have appeared that need to be addressed.

APPENDIX D (b):

CONSTRUCTABILITY PROCEDURE – SIGNIFICANCE OF BARRIER

Significance of Barrier			General Organization Barriers
Insignificant	Significant	Very Significant	
☐	☐	☐	• Complacency with the status quo - resistance to change - conservative, non-innovative approaches - risk-averse attitudes towards trying something new - no rewards for intelligent risk-taking - a "not invented here" syndrome
☐	☐	☐	• Lack of documentation and retrieval of "lessons learned" - no formal system for documenting lessons learned - reliance on work-of-mouth and experienced personnel to transfer innovative ideas
☐	☐	☐	• The "right people" were / are not available
☐	☐	☐	Owner Project Barriers • Lack of awareness / understanding of the concepts of constructability; no procedural "roadmap" is available - constructability used as a buzzword - efforts ineffective due to lack of coordination, direction
☐	☐	☐	• Perception that "we do it"; very narrow view of constructability - "routine design practices fully exploit constructability" - "we already pay for it" - "We do value engineering; value engineering equals constructability"
☐	☐	☐	• Lack of team-building or partnering - client-contractor relationships / communications are not respected and nurtured - adversarial relationships are free to develop (expected, accepted, and perhaps even subconsciously promoted)
☐	☐	☐	• Lack of documentation and retrieval of "lessons learned" - no formal system for documenting lessons learned - reliance on work-of-mouth and experienced personnel to transfer innovative ideas
☐	☐	☐	• Lack of genuine commitment to constructability - constructability is a low priority - no policy statement exists, no champion - "There are higher priorities"; "There are bigger fish"

APPENDIX D (c): CONSTRUCTABILITY PROCEDURE –BARRIER INPUT

Barrier	Results (CALCULATED)			Number of Responses (INPUT)		
	Insignificant	Significant	Very Significant	Insignificant	Significant	Very Significant
1-Complacency with the status quo	0	0	0			
2-Lack of documentation and retrieval of "lessons learned"	0	0	0			
3-The "right people" were / are not available	0	0	0			
4-Lack of awareness / understanding of the concepts of constructability; no procedural "roadmap" is available	0	0	0			
5-Perception that "we do it"; very narrow view of constructability	0	0	0			
6-Lack of team-building or partnering	0	0	0			
7-Lack of documentation and retrieval of "lessons learned"	0	0	0			
8-Lack of genuine commitment to constructability	0	0	0			
9-There are no proven benefits of constructability	0	0	0			
10-Misdirected design objectives and designer performance measures	0	0	0			
11-Use of lump-sum competitive contracting, leading to:	0	0	0			
12-Reluctance to invest additional money, effort, and time in early project stages	0	0	0			
13-Perception that "we do it"; very narrow view of constructability	0	0	0			
14-Lack of awareness / understanding of constructability concepts and/or benefits	0	0	0			
15-Lack of construction experience / qualified personnel	0	0	0			
16-Lack of mutual respect between designers and constructors	0	0	0			
17-Poor timeliness of input	0	0	0			
18-Contractor or construction input is requested too late to be of value	0	0	0			
19-Poor communication skills; design criticism is often non-constructive or communicated in a offensive, tactless	0	0	0			
20-Other	0	0	0			
21-Other	0	0	0			
Weightings	1	3	5			
DATA INPUT AREAS						

Link to [Constructability Procedure – Barrier Input Sheet](#)

**APPENDIX D (d):
CONSTRUCTABILITY PROCEDURE – BARRIER ASSESSMENT RESULTS**

The listing of barrier breakers presents methods for overcoming the seven most prevalent barriers to constructability. **Appendix G** presents the barrier breakers in the form of a reference chart that also shows where the barrier breaker should be applied, and how the barrier breaker affects the barrier.

This tool supplements Appendices C-F. Once barriers have been identified, corporate or project staffs may use these high-impact strategies as a starting point for mitigation efforts. In addition to being effective tactics for addressing particular barriers, these barriers breakers may serve as examples for identifying additional effective methods to overcome barriers.

APPENDIX D (e):

CONSTRUCTABILITY PROCEDURE – BARRIER BREAKERS

Barrier	Recommended Barrier Breakers	Where Effective		Type of Breaker				Roadmap Activities
		Corporate	Project	Cultural	Procedural	Awareness	Incentive	
1. Complacency with status quo.	• Designate a strong program champion							"Identify constructability sponsor / champion"
2. Reluctance to invest additional money and effort in early project stages.	• Promote the attitude that constructability should be viewed as an investment opportunity with corresponding downstream payoff.							"Assess and recognize constructability benefits" "Define constructability objectives and measures"
	• Include constructability as part of a standard bid response and in cost tracking / control efforts.							"Select project contracting strategy"
3. Limitations of lump-sum competitive contracting.	• Owner / designer acquire in-house construction expertise for input during design.							"Assemble key owner team members" "Develop the constructability team"
	• Develop a short list of contractors who offer constructability input in return for the opportunity to be on the short list of bidders.							"Select project contracting strategy" "Secure contractors, vendors, and consultants"
4. Lack of construction experience in design organization.	• Communicate construction issues from field engineers to office engineers / designers.							"Consult applications matrix and lessons learned"
	• Close the "project loop" by getting feedback from the field and by tracking lessons learned.							"Document lessons learned"
	• Modify design management practices to elevate the visibility of constructability issues.							"Develop implementation policy" "Define constructability objectives and measures"
5. Designer's perception that "we do it."	• Secure comprehensive understanding of what constructability is prior to assessing adequacy of efforts.							"Understand constructability objectives, methods, concepts, and barriers"
6. Lack of mutual respect between designers and constructors.	• Aggressively promote effective team-building among project personnel.							"Develop the constructability team"
	• Establish constructor presence in design process before pride of authorship develops.							"Develop constructability procedures and integrate into project activities" "Select project contracting strategy"
	• Keep the project team focused on common objectives and accepted procedures rather than personalities.							"Orient project constructability team / perform team-building"
7. Construction input is requested too late to be of value.	• Increase awareness of the necessity for early construction involvement.							"Understand constructability objectives, methods, concepts, and barriers"
	• Include constructability as an early activity in a formal project activity flow plan or roadmap.							All Roadmap Activities
	• Include individuals with significant construction experience in the project team from the outset.							"Assemble key owner team members" "Select project contracting strategy"
Sum of effective breakers in each category:								

Link to [Constructability Procedure – Barrier Breakers](#)

APPENDIX E: CONSTRUCTABILITY PROCEDURE – WORKSHOP AGENDA

- I. Introductions
- II. Safety Discussion
- III. Review of Agenda
- IV. Project Status Briefing
- V. Results/Overview of Constructability Orientation and Assessment of Barriers to Implementation
- VI. Small Group Breakout Sessions: Identification of Constructability Concerns and Opportunities
- VII. Reports from Concerns / Opportunities Break-Out Sessions
- VIII. Prioritization of Constructability Concepts and Sub-Concepts
- IX. Results of Pre-Screening of Lessons Learned, Plans for Further Screening
- X. Small Group Breakout Sessions: Development of Concept Application Plans
- XI. Reports from Concept Application Plans Breakout Sessions
- XII. Path Forward for Implementing Concept Application Plans
- XIII. Close

APPENDIX F:
CONSTRUCTABILITY PROCEDURE – SUGGESTION LOG

Unique Tracking Number	Constructability Suggestion Description	Originator	Work Package	Discipline	Sub-discipline level	Supplier Site or Fabrication / Construction Site	Perceived Benefit	Cost benefit	Safety benefit	Date Suggestion Made	Proposed Solution	Individual Responsible for Implementation	Target Implementation Project Phase	Target Implementation Date (Execute Phase)	Schedule Impact / Decision milestone required	Action Plan	Progress	Final Disposition Date	Did we get the results expected?
001																			
002																			
003																			
004																			
005																			
006																			
007																			
008																			
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010																			
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038																			
039																			
040																			

Link to [Constructability Procedure - Suggestion Log Sheet](#)

**APPENDIX G:
CONSTRUCTABILITY PROCEDURE – STUDY FORMAT**

- A.** Concept Targeted: _____
- B.** Concerns / drivers associated with the Concept: _____

- C.** Opportunities / drivers associated with the Concept: _____

- D.** Decision to be Impacted / Supported: _____
- E.** Deliverables Needed: _____
- F.** Purpose of deliverables: _____
- G.** Sub-components of deliverables: _____
- H.** Targeted customer of user of deliverables: _____

- I.** Needed timing of completion of deliverables and associated timing constraints: _____

- J.** Recommended lead individual responsible for completing deliverables: _____

- K.** Recommended format of deliverables: _____
- L.** Individuals to be consulted in preparing these deliverables: _____

- M.** Additional key information sources for developing these deliverables: _____

**APPENDIX H:
CONSTRUCTABILITY PROCEDURE – SUGGESTION FORM**

Topic: _____

Discipline(s) / Craft(s) Affected:

Description and Illustration:

Originated by: _____ **Date:** _____

- Assessment of Impact to Project:** (to be completed with the assistance of the Constructability Coordinator)
- Costs:
 - Schedule:
 - Quality:
 - Safety:
 - Engineering / Procurement:
 - Need to change / update corporate standard specs?
 - Other:

Approvals:

Comments:

**APPENDIX I:
CONSTRUCTABILITY PROCEDURE – EVALUATION MATRIX**

This form shows the 14 critical parameters identified for assessing constructability implementation. These parameters have been defined by a five-level program classification. Level 5 is considered the highest level of implementation of constructability.

To perform the self-assessment, distribute the Evaluation Matrix to a variety of experienced personnel throughout the organization. Separate assessments from different organizational levels should be solicited for the evaluation to be effective. Within the boxes, each participant should mark the level at which he or she feels the organization is operating for each of the 14 parameters.

A review of the evaluation from all personnel surveyed will allow the Constructability champion, to determine areas where the constructability program is successful, and also to identify areas where improvement is required. A baseline number for comparison and evaluation may be obtained using a simple average of the rankings for the parameters. The 14 parameters are each given equal weight, as all are of paramount importance for effective constructability implementation.

An initial assessment provides a baseline for program improvement. Periodically, the Evaluation Matrix should be used to reassess constructability procedures for effectiveness.

Program Classification

1 No Program	2 Application of Selected Supports	3 Informal Program	4 Formal Program	5 Comprehensive Formal Program
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I: Corporate Culture

A. Program Designation	No designation	Part of standard construction management activities	Part of another program, such as Quality, or only identified on a project level	Recognized on a corporate level, but may be part of another program	Stand-alone program on same level as Quality or Safety
B. Management Attitude Toward Constructability	No recognition of need for or benefits of constructability	Limited support within company; some recognition of benefits	Support varies within company; support on a project-by-project basis	Management supports program; constructability understood as a corporate philosophy	Total management support; actively participate in improving program
C. Recognition of Constructability Barriers/Problems	Many barriers exist; no recognition of barriers or problems encountered	Many barriers exist; aware of external limiters; may deny internal factors	Recognize presence of barriers/problems; accept as part of the job	Actively identify, work to document and correct	Most barriers gone; problems caught and corrected quickly
D. Constructability Training of Personnel	None	If any occurs, done as part of on-the-job training	Awareness seminars for specific projects	Part of standard orientation	Part of standard orientation; deeply ingrained in corporate culture

II: Personnel

A. Executive Sponsor for Constructability	No sponsor	No sponsor	Sponsor identified; role of sponsor may be ambiguous or weak	Yes; sponsor actively supports program	Yes; sponsor actively supports program
B. Assignment of Corporate Constructability Duties	None	No corporate personnel dedicated to constructability	May have corporate duties as part of other responsibilities	Full or part-time corporate coordinator; corporate support organization for program implementation	Full-time, high-level corporate coordinator; support organization well developed
C. Role of Project Constructability Coordinator	Not identified	Part-time if identified; very limited responsibility	Full or part-time position; responsibilities vary by project size, type, participants	Full or part-time position; responsibilities vary by project size, type, participants	Full-time position; plays major project role

III: Documentation/Tracking

A. Constructability Program Documentation	None	Limited reference in any manuals; CII documents may be distributed or referenced	Project-level program documents exist; may be included in other corporate documents	Corporate constructability manual is available	Corporate constructability manual is thorough, widely distributed, and periodically updated
B. Tracking of Constructability Lessons Learned	None	Believe that ideas are adequately conveyed via work of mouth, personnel interaction	Some individual documentation; primarily post project reviews and reports	System exists for capture and communication of lessons learned	Database on lessons learned involves input from all levels
C. Sharing Advanced Construction Technologies	Not done	New information routed occasionally – journals, word-of-mouth	Library may exist; information routinely routed or seminars held	Formalized routing system; R&D department identifies and promotes	Formalized system with company seminars and pilot applications
D. Constructability Referenced in Contract Documents	No reference	Limited reference, on specific projects; often only at request of other project participants	Level of reference varies by project type, role, or participants	Standard item in all contracts	Standard item in all contracts; actively promoted to other organizations
E. Tracking Constructability Savings/Effects	Not applicable	No tracking or recognition of program results	No tracking, limited recognition of program results on project	Track for particular projects or selected items; may track major ideas across project	Data kept on all projects; widespread confidence in sayings beyond those measured

IV: Implementation

A. Nature of Project-level Efforts & Inputs	None	Reactive approach, constrained by review mentality, lack of understanding of proactive benefits	Aware of major benefits proactive approach; efforts vary project by project	Proactive effort on all projects; routinely consult lessons learned	Aggressive, proactive efforts from beginning of project; routinely consult lessons learned
B. Implementation of Constructability Concepts	None	Some concepts used periodically; often considered too late to be of use	Selected concepts applied regularly; full use, timeliness of input varies across projects	All concepts consistently considered; timely implementation of feasible concepts	All concepts consistently considered, continuously evaluated, aggressively implemented

APPENDIX J:
CONSTRUCTABILITY PROCEDURE – DISCIPLINE CHECKLISTS

Link to [Constructability Procedure - Discipline Checklists sheets](#)

APPENDIX K:
CONSTRUCTABILITY PROCEDURE – FLOWCHART OF ACTIVITIES

